Prototype Planning & AI Studio Implementation Strategy

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Field	Details
Project Name	ResumeAI — Smart Resume Upscaler
Prototype Type	Functional web prototype (React + TypeScript + Firebase + Gemini API)
AI Tool	Google Gemini 1.5 Pro via AI Studio API
Tech Stack	React, TypeScript, Firebase Auth, Firestore, Vite, Tailwind CSS
Status	Prototype built and running — UI live, AI integration functional

01 Prototype Overview

The ResumeAI prototype is a fully functional web application built to validate the core concept before moving to a native Android implementation via Android Studio. The prototype demonstrates that the AI-powered resume analysis workflow is technically feasible and delivers genuine value to users. It serves as the proof of concept that will guide the structure of the final mobile app.

The application is powered by Google's Gemini 1.5 Pro model accessed through AI Studio's API. Users can paste their resume text, receive an ATS compatibility score, get keyword gap feedback, and have their bullet points rewritten by AI — all within a clean, modern interface.

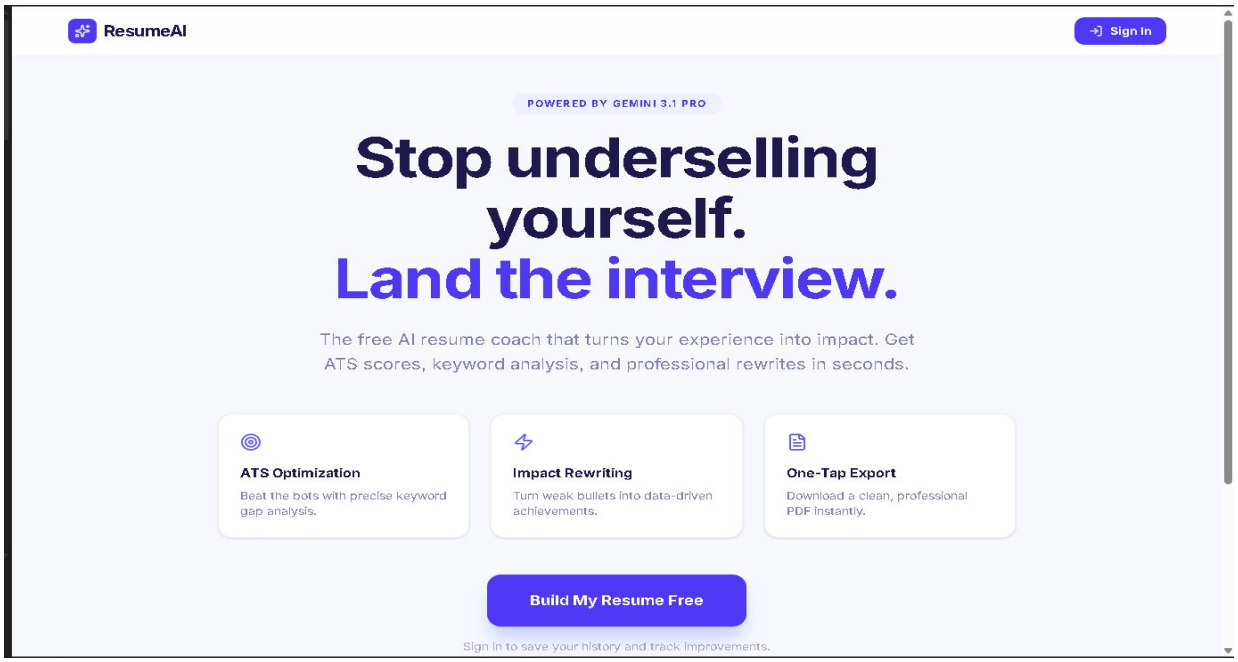

Fig 1: ResumeAI Prototype — Landing Page (Powered by Gemini 1.5 Pro)

02 Key Features in the Prototype

ATS Optimization	Analyzes resume text for keyword density and ATS compatibility. Highlights missing keywords relative to target job descriptions.
Impact Rewriting	AI rewrites weak or vague bullet points into strong, quantified, action-driven achievement statements.
One-Tap PDF Export	Generates a clean, professionally formatted PDF of the improved resume instantly from the browser.
Google Authentication	Firebase Auth with Google Sign-In so users can save their resume history and track improvements across sessions.
Firestore Storage	User resume data and improvement history stored securely in Firestore for persistence across sessions.

03 Role of AI Studio (Gemini API)

AI Studio was used to design, test, and deploy the prompt engineering layer that powers ResumeAI's intelligence. The Gemini 1.5 Pro model was selected for its strong performance on structured text analysis tasks and its ability to return consistent, parseable JSON responses.

How AI Studio Was Used:

- Prompt design and iteration — tested resume analysis prompts in the AI Studio playground before integrating them into code
- Structured output — prompts engineered to return JSON with score, missing keywords, and rewritten bullets
- API key generation — Gemini API key obtained from AI Studio and secured in environment variables
- Model selection — evaluated Gemini 1.5 Pro vs Flash for quality vs latency tradeoff
- Edge case testing — tested prompts against varied resume formats and job descriptions

Core Prompt Structure:

```
You are an expert resume coach and ATS specialist. Analyze the following resume against
the job description provided. Return a JSON object with: - ats_score: number (0-100) -
missing_keywords: string[] - improved_bullets: { original: string, improved: string }[]
- section_feedback: { section: string, feedback: string }[]
```

04 Tech Stack & Code Architecture

The prototype was built as a React + TypeScript web app using Vite for fast development builds. The codebase is cleanly separated into components and services — making it easy to lift the core AI logic later into a Flutter/Android Studio implementation.

File / Module	Purpose
geminiService.ts	Handles all Gemini API calls — sends resume + job desc, parses JSON response
ResumeForm.tsx	Input form for resume text and job description paste
AnalysisView.tsx	Displays ATS score, keyword gaps, and rewritten bullets
PDFExport.ts	Generates and downloads the improved resume as PDF
firebase.ts	Firebase config — Auth (Google Sign-In) + Firestore setup
App.tsx	Root component — manages auth state and screen routing

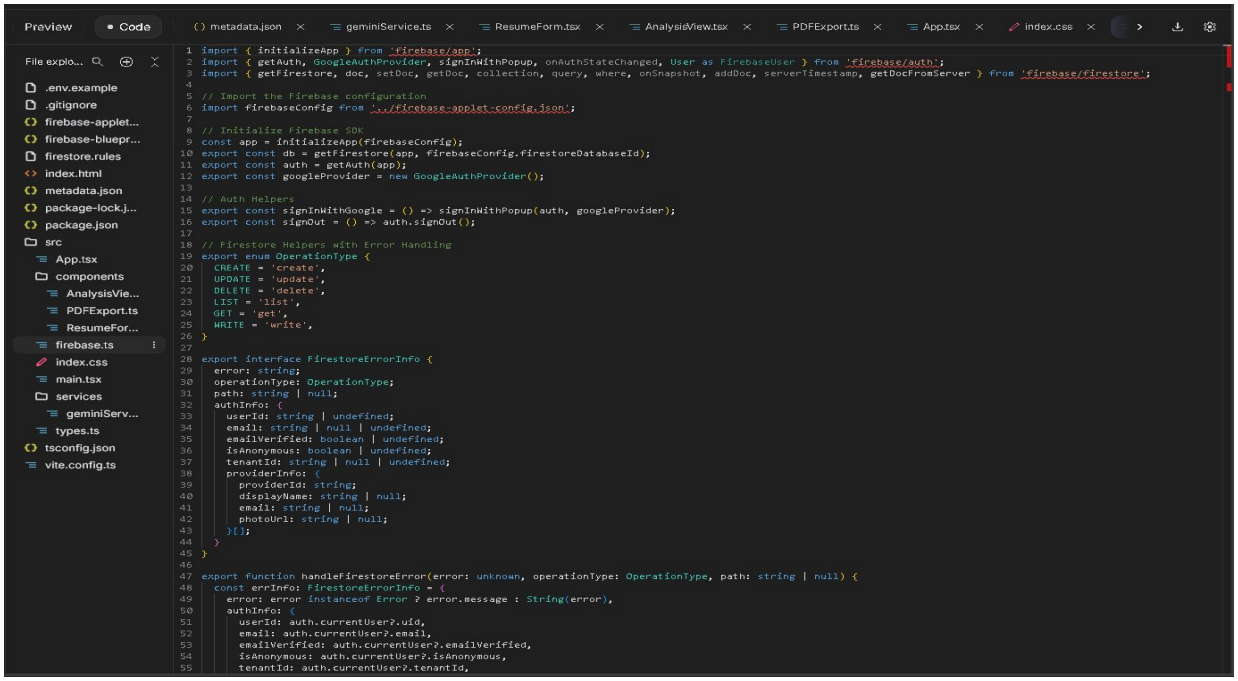

Fig 2: ResumeAI Codebase — firebase.ts showing Firebase Auth, Firestore & Gemini integration

05 Android Studio Implementation Plan

The web prototype validates the complete AI workflow. The next phase moves this into a native Android app using Flutter in Android Studio, translating each component of the prototype into its mobile equivalent.

Prototype Component	Android Studio / Flutter Equivalent
React UI components	Flutter widgets — Column, TextField, Card, ElevatedButton
geminiService.ts	Dart service class using http package to call Gemini REST API
Firebase Auth (Google)	firebase_auth + google_sign_in Flutter packages
Firestore storage	cloud_firestore Flutter package — same data model
PDF Export	pdf + printing Flutter packages for on-device PDF generation
Vite dev server	Android Studio emulator + physical device testing via ADB

06 How the Prototype Validates the Idea

- The AI analysis flow works end-to-end — resume in, structured feedback out, in under 3 seconds
- Gemini's JSON output is consistent and parseable, confirming the API is reliable enough for production
- Google Auth and Firestore integration confirms the user session and history management approach is solid
- The landing page and UI flow validates that the user experience is intuitive without onboarding
- PDF export confirms the output delivery mechanism works before rebuilding it in Flutter

Next Step	With the prototype confirming the full AI workflow, the implementation will move into Android Studio — rebuilding the same experience as a native Flutter app for mobile deployment.